

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

1. NAME OF THE MEDICINAL PRODUCT

ZYVASTA 2

Pitavastatin Film-Coated Tablets 2mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film coated tablet contains:

Pitavastatin calcium equivalent to

Pitavastatin

2mg

Colouring: Titanium Dioxide

For a full list of excipients see section 6.1.

3. PHARMACEUTICAL FORM

White to off white, beveled edge, biconvex round shaped film coated tablets plain on both sides. The tablets should be free of all physical defects.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Pitavastatin is indicated for the reduction of elevated total cholesterol (TC) and LDL-C, in adults, adolescents and children aged 6 years or older with primary hypercholesterolaemia, including heterozygous familial hypercholesterolaemia, and combined (mixed) dyslipidaemia, when response to diet and other nonpharmacological measures is inadequate.

4.2 Posology and method of administration

Posology

Patients should be on a cholesterol lowering diet before treatment. It is important that all patients continue dietary control during treatment. The usual starting dose is 1mg or 2mg once daily. Adjustment of dose should be made at intervals of 4 weeks or more. Doses should be individualized according to LDL-C levels, the goal of therapy and patient response. The maximum daily dose is 4mg.

Elderly

No dose adjustment necessary in patients older than 70 years.

Paediatric population

Children and adolescents aged 6 years and over:

Pitavastatin use in children should only be carried out by physicians experienced in the treatment of hyperlipidaemia and progress should be regularly reviewed.

In children and adolescents with heterozygous familial hypercholesterolaemia the usual starting dose is 1mg once daily. Adjustment of dose should be made at intervals of 4 weeks or more. Doses should be individualized according to LDL-C levels, the goal of therapy and patient response. In children 6 to 9 years of age the maximum daily dose is 2mg. In children 10 years or older the maximum daily dose is 4mg.

Children younger than 6 years of age:

The safety and efficacy of pitavastatin in children below 6 years of age has not been established and no data are available.

Patients with impaired renal function

Moderate and severe renal impairment (glomerular filtration rate 30 – 59 and 15 – 29mL/min/1.73m², respectively) as well as end-stage renal disease on hemodialysis: Starting dose of 1mg once daily and maximum dose of 2mg once daily.

Patients with mild to moderate impaired hepatic function

The 4mg dose is not recommended in patients with mild to moderate impaired hepatic function. A maximum daily dose of 2mg may be given with close monitoring.

Method of Administration

For oral use only and should be swallowed whole. Pitavastatin can be taken at any time of the day with or without food. It is desirable that the patient takes the tablet at the same time each day. Statin therapy is generally more effective in the evening due to the circadian rhythm of lipid metabolism.

If a child or adolescent is unable to swallow the tablet, where necessary, the tablet may be dispersed in a glass of water and taken immediately. To ensure accurate dosage a second volume of water should be used to rinse the glass and swallowed immediately. Tablets must not be dispersed in either acidic fruit juices or milk.

Zyvasta 2 is available at the strength of 2mg only and may not be able to deliver all the dosing recommendations mentioned above. In such cases, other approved strengths of pitavastatin should be used.

4.3 Contraindications

Pitavastatin is contraindicated:

- in patients with known hypersensitivity to pitavastatin or to any of the excipients or other statins
- in patients with severe hepatic impairment, active liver disease or unexplained persistent elevations in serum transaminases (exceeding 3 times the upper limit of normal [ULN])
- in patients with myopathy
- in patients receiving concomitant ciclosporin
- during pregnancy, while breast feeding and in women of child bearing potential not taking appropriate contraceptive precautions

4.4 Special warnings and precautions for use

Muscle Effects: In common with other HMG-CoA reductase inhibitors (statins), there is the potential for myalgia, myopathy and, rarely, rhabdomyolysis to develop. Patients should be asked to report any muscle symptoms. Creatine kinase (CK) levels should be measured in any patient reporting muscle pain, muscle tenderness or weakness especially if accompanied by malaise or fever.

Creatine kinase should not be measured following strenuous exercise or in the presence of any other plausible cause of CK increase which may confound interpretation of the result. When elevated CK concentrations (>5x ULN) are noted, a confirmatory test should be performed within 5 to 7 days. There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

Pitavastatin must not be co-administered with systemic formulations of fusidic acid or within 7 days of stopping fusidic acid treatment. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving fusidic acid and statins in combination. The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness. Statin therapy may be re-introduced seven days after the last dose of fusidic acid. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g., for the treatment of severe infections, the need for co-administration of pitavastatin and fusidic acid should only be considered on a case by case basis and under close medical supervision.

Before Treatment

In common with other statins, Pitavastatin should be prescribed with caution in patients with pre-disposing factors for rhabdomyolysis. A creatinine kinase level should be measured, to establish a reference baseline, in the following situations:

- renal impairment,
- hypothyroidism,
- personal or family history of hereditary muscular disorders,
- previous history of muscular toxicity with a fibrate or another statin,
- history of liver disease or alcohol abuse,
- elderly patients (over 70 years) with other predisposing risk factors for rhabdomyolysis,

In such situations, clinical monitoring is recommended and the risk of treatment should be considered in relation to the possible benefit. Treatment with pitavastatin should not be started if CK values are >5x ULN.

During Treatment

Patients must be encouraged to report muscle pain, weakness or cramps immediately. Creatine kinase levels should be measured and treatment stopped if CK levels are elevated (>5x ULN). Stopping treatment should be considered if muscular symptoms are severe even if CK levels are ≤5x ULN. If symptoms resolve and CK levels return to normal, then re-introduction of pitavastatin may be considered at a dose of 1mg and with close monitoring.

Liver Effects: In common with other statins, Pitavastatin should be used with caution in patients with a history of liver disease or who regularly consume excessive quantities of alcohol. Liver function tests should be performed prior to initiating treatment with pitavastatin and then periodically during treatment. Pitavastatin treatment should be discontinued in patients who have a persistent increase in serum transaminases (ALT and AST) exceeding 3x ULN.

Renal Effects: Pitavastatin should be used with caution in patients with moderate or severe renal impairment. Dose increments should be instituted only with close monitoring. In those with severe renal impairment, 4mg dose is not recommended.

Diabetes Mellitus: Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk of hyperglycaemia (fasting glucose 5.6 to 6.9mmol/L, BMI>30kg/m², raised triglycerides, hypertension), should be monitored both clinically and biochemically according to national guidelines. However, there has been no confirmed signal of a diabetes risk for pitavastatin either in post-marketing safety surveillance studies or in prospective studies.

Interstitial Lung Disease: Exceptional cases of interstitial lung disease have been reported with some statins, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Paediatric population: There is limited data on the long term effect on growth and sexual maturation in paediatric patients 6 years of age or older taking pitavastatin. Adolescent females should be counselled on appropriate contraceptive precautions during treatment with pitavastatin.

Other effects: A temporary suspension of pitavastatin is recommended for the duration of treatment with erythromycin, other macrolide antibiotics or fusidic acid. Pitavastatin should be used with caution in patients taking drugs known to cause myopathy (e.g. fibrates or niacin). The tablets contain lactose. Patients with the rare hereditary problems of galactose intolerance, Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine. In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Pitavastatin should be discontinued in case of aggravation of symptoms.

Recurrences when the same or a different statin was (re-) administered have been reported.

4.5 Interaction with other medicinal products and other forms of interaction

Pitavastatin is actively transported into human hepatocytes by multiple hepatic transporters (including organic anion transporting polypeptide, OATP), which may be involved in some of the following interactions.

Ciclosporin: Co-administration of a single dose of ciclosporin with pitavastatin at steady state resulted in a 4.6-fold increase in pitavastatin AUC. The effect of steady state ciclosporin on steady state pitavastatin is not known. Pitavastatin is contraindicated in patients being treated with ciclosporin.

Erythromycin: Co-administration with pitavastatin resulted in a 2.8-fold increase in pitavastatin AUC. A temporary suspension of pitavastatin is recommended for the duration of treatment with erythromycin or other macrolide antibiotics.

Gemfibrozil and other fibrates: The use of fibrates alone is occasionally associated with myopathy. Co-administration of fibrates with statins may cause severe myositis and myoglobinuria, and has been associated with increased myopathy and rhabdomyolysis. Pitavastatin should be administered with caution when used concomitantly with fibrates. In pharmacokinetic studies co-administration of pitavastatin with Gemfibrozil resulted in a 1.4-fold increase in pitavastatin AUC; with Fenofibrate, AUC increased 1.2-fold.

Niacin: Interaction studies with pitavastatin and niacin have not been conducted. The use of niacin alone has been associated with myopathy and rhabdomyolysis when used as a monotherapy. Thus pitavastatin should be administered with caution when used concomitantly with niacin.

Fusidic acid: The risk of myopathy including rhabdomyolysis may be increased by the concomitant administration of systemic fusidic acid with statins. The mechanism of this interaction (whether it is pharmacodynamic or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. If treatment with systemic fusidic acid is necessary, pitavastatin treatment should be discontinued throughout the duration of the fusidic acid treatment.

Rifampicin: Co-administration with pitavastatin at the same time resulted in a 1.3-fold increase in pitavastatin AUC due to reduced hepatic uptake.

Protease inhibitors and Non-nucleoside reverse transcriptase inhibitors: Co-administration of lopinavir/ritonavir, darunavir/ritonavir, atazanavir, or efavirenz with pitavastatin at the same time may result in minor changes in pitavastatin AUC.

Ezetimibe and its glucuronide metabolite inhibit the absorption of dietary and biliary cholesterol. Coadministration of pitavastatin had no effect on plasma ezetimibe or the glucuronide metabolite concentrations and ezetimibe had no impact on pitavastatin plasma concentrations.

Inhibitors of CYP3A4: Interaction studies with itraconazole and grapefruit juice, known inhibitors of CYP3A4, had no clinically significant effect on the plasma concentrations of pitavastatin. **Digoxin**, a known P-gp substrate, did not interact with pitavastatin. During co-administration there was no significant change in either pitavastatin or digoxin concentrations.

Warfarin: The steady-state pharmacokinetics and pharmacodynamics (INR and PT) of warfarin in healthy volunteers was unaffected by the co-administration of pitavastatin 4mg daily. However, as for other statins, patients receiving warfarin should have their prothrombin time or INR monitored when pitavastatin is added to their therapy.

Paediatric population: Drug-drug interaction studies have only been performed in adults. The extent of interactions in the paediatric population is not known.

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4.6 Fertility, pregnancy and lactation

Pregnancy: Pitavastatin is contraindicated during pregnancy. Women of childbearing potential must take appropriate contraceptive precautions during treatment with pitavastatin. Since cholesterol and other products of cholesterol biosynthesis are essential for the development of the fetus, the potential risk for inhibition of HMG-CoA reductase outweighs the advantage of treatment during pregnancy. Animal studies show evidence of reproductive toxicity, but no teratogenic potential. If the patient is planning to become pregnant, treatment should be stopped at least one month prior to conception. If a patient becomes pregnant during use of pitavastatin, treatment must be discontinued immediately.

Breastfeeding: Pitavastatin is contraindicated during breastfeeding. Pitavastatin is excreted in rat milk. It is not known whether it is excreted in human milk.

Fertility: No current data

4.7 Effects on ability to drive and use machines

There is no pattern of adverse events that suggests that patients taking pitavastatin will have any impairment of ability to drive and use hazardous machinery, but it should be taken into account that there have been reports of dizziness and somnolence during treatment with pitavastatin.

4.8 Undesirable effects

Summary of adverse reactions: Adverse reactions and frequencies observed in worldwide controlled clinical trials and extension studies, at the recommended doses, are listed below by system organ class.

Frequencies are defined as: very common (≥1/10), common (≥1/100, to <1/10), uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1,000) very rare (<1/10,000) and not known.

Blood and the lymphatic system disorders

Uncommon: Anaemia

Metabolism and nutrition disorders

Uncommon: Anorexia

Psychiatric disorders

Uncommon: Insomnia

Nervous system disorders

Common: Headache, Uncommon: Dizziness, Dysgeusia, Somnolence, *not known:* Myasthenia gravis

Eye disorders

Rare: Visual acuity reduced, *not known:* Ocular myasthenia

Ear and labyrinth disorders

Uncommon: Tinnitus

Gastrointestinal disorders

Common: Constipation, Diarrhoea, Dyspepsia, Nausea, Uncommon: Abdominal Pain, Dry Mouth, Vomiting,

Rare: Glossodynia, pancreatitis acute

Hepato-biliary disorders

Uncommon: Transaminases (aspartate aminotransferase, alanine aminotransferase) increased, Rare: Jaundice cholestatic

Skin and subcutaneous tissue disorders

Uncommon: Pruritus, Rash, Rare: Urticaria, Erythema

Musculoskeletal, connective tissue and bone disorders

Common: Myalgia, Arthralgia, Uncommon: Muscle spasms, Frequency unknown: Immune-mediated necrotising myopathy.

Renal and urinary disorders

Uncommon: Pollakiuria

General disorders and administration site conditions

Uncommon: Asthenia, Malaise, Fatigue, Peripheral Oedema

Paediatric population:

The most commonly reported pitavastatin related adverse reactions in the clinical programme were headache (4.9%), myalgia (2.1%) and abdominal pain (4.9%). Based on the data available, the frequency, type and severity of adverse reactions are expected to be similar in children and adolescents to adults.

Adverse reactions and frequencies observed in the prospective post-marketing surveillance study but not in worldwide controlled clinical trials, at the recommended doses are listed below.

Hepato-biliary disorders

Rare: Hepatic function abnormal, Liver disorder

Musculoskeletal, connective tissue disorders

Rare: Myopathy, Rhabdomyolysis

In the post-marketing surveillance study there were two reports of rhabdomyolysis requiring hospitalisation (0.01% of patients).

In addition, there are unsolicited post-marketing reports of skeletal muscle effects including myalgia and myopathy in pitavastatin treated patients at all recommended doses. Reports of rhabdomyolysis, with and without acute renal failure, including fatal rhabdomyolysis have also been received. Unsolicited reports of the following events have also been received (the frequency is based on that observed in post-marketing studies).

Nervous system disorders

Uncommon: Hypoaesthesia

Gastrointestinal disorders

Rare: Abdominal discomfort

Statin class effects: The following adverse events have been reported with some statins:

- Sleep disturbances, including nightmares
- Cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion)
- Sexual dysfunction
- Depression
- Exceptional cases of interstitial lung disease, especially with long term therapy
- Diabetes Mellitus: Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥5.6mmol/L, BMI >30kg/m2, raised triglycerides, history of hypertension)
- Hyperglycemia
- Increase in HbA1c

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks).

Increases in HbA1c and fasting blood glucose have been reported with statins.

The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Musculoskeletal disorders:

Frequency not known: Immune-mediated necrotizing myopathy

4.9 Overdose

There is no specific treatment in the event of overdose. The patient should be treated symptomatically and supportive measures instituted as required. Liver function and CK levels should be monitored.

Haemodialysis is unlikely to be of benefit.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: HMG-CoA reductase inhibitors. ATC Code: C10A A08

Mechanism of Action: Pitavastatin competitively inhibits HMG-CoA reductase, the rate-limiting enzyme in the biosynthesis of cholesterol, and inhibits cholesterol synthesis in the liver. As a result the expression of LDL receptors in the liver is increased, promoting the uptake of circulating LDL from the blood, decreasing total cholesterol (TC) and LDL-cholesterol (LDL-C) concentrations in the blood. Its sustained inhibition of hepatic cholesterol synthesis reduces VLDL secretion into the blood, reducing plasma triglyceride (TG) levels.

Pharmacodynamic Effects: Pitavastatin reduces elevated LDL-C, total cholesterol and triglycerides and increases HDL-cholesterol (HDL-C). It reduces Apo-B, and produces variable increases in Apo-A1. It also reduces non-HDL-C and elevated TC/HDL-C, and Apo-B/Apo-A1 ratios.

Table 1: Dose response in patients with primary hypercholesterolaemia (Adjusted mean percent change from baseline over 12 weeks)

Dose	N	LDL-C	TC*	HDL-C	TG	Apo-B	Apo-A1
Placebo	51	-4.0	-1.3	2.5	-2.1	0.3	3.2
1mg	52	-33.3	-22.8	9.4	-14.8	-24.1	8.5
2mg	49	-38.2	-26.1	9.0	-17.4	-30.4	5.6
4mg	50	-46.5	-32.5	8.3	-21.2	-36.1	4.7

*unadjusted

5.2 Pharmacokinetic properties

Absorption: Pitavastatin is rapidly absorbed from the upper gastrointestinal tract and peak plasma concentrations are achieved within one hour after oral administration. Absorption is not affected by food. Unchanged drug undergoes enterohepatic circulation and is well absorbed from the jejunum and ileum. The absolute bioavailability of pitavastatin is 51%.

Distribution: Pitavastatin is more than 99% protein bound in human plasma, mainly to albumin and alpha 1-acid glycoprotein, and the mean volume of distribution is approximately 133L. Pitavastatin is actively transported into hepatocytes, the site of action and metabolism, by multiple hepatic transporters including OATP1B1 and OATP1B3. Plasma AUC is variable with an approximately 4-fold range between the highest and lowest values. Studies with SLCO1B1 (the gene which encodes OATP1B1) suggests that polymorphism of this gene could account for much of the variability in AUC. Pitavastatin is not a substrate for p-glycoprotein.

Biotransformation: Unchanged pitavastatin is the predominant drug moiety in plasma. The principal metabolite is the inactive lactone which is formed via an ester-type pitavastatin glucuronide conjugate by UDP glucuronosyltransferase (UGT1A3 and 2B7). In vitro studies, using 13 human cytochrome P450 (CYP) isoforms, indicate that the metabolism of pitavastatin by CYP is minimal; CYP2C9 (and to a lesser extent CYP2C8) is responsible for the metabolism of pitavastatin to minor metabolites.

Elimination: Unchanged pitavastatin is rapidly cleared from the liver in the bile, but undergoes enterohepatic recirculation, contributing to its duration of action. Less than 5% of pitavastatin is excreted in the urine. The plasma elimination half-life ranges from 5.7 hours (single dose) to 8.9 hours (steady state) and the apparent geometric mean oral clearance is 43.4L/h after single dose.

Effect of food: The maximum plasma concentration of pitavastatin was reduced by 43% when it was taken with a high-fat meal, but AUC was unchanged.

Special populations

Elderly: It was reported in a pharmacokinetic study which compared healthy young and elderly (≥65 years) volunteers, pitavastatin AUC was 1.3-fold higher in elderly subjects. This has no effect on the safety or efficacy of pitavastatin in elderly patients.

Gender: It was reported in a pharmacokinetic study which compared healthy male and female volunteers, pitavastatin AUC was increased 1.6-fold in women. This has no effect on the safety or efficacy of pitavastatin in women.

Race: There was no difference in the pharmacokinetic profile of pitavastatin between Japanese and Caucasian healthy volunteers when age and body weight was taken into account.

Paediatric population: There are limited pharmacokinetic data in children and adolescents.

Renal insufficiency: For patients with moderate renal disease and those on haemodialysis increases in AUC values were 1.8-fold and 1.7-fold respectively.

Hepatic insufficiency: For patients with mild (Child-Pugh A) hepatic impairment AUC was 1.6 times that in healthy subjects, while for patients with moderate (Child-Pugh B) hepatic impairment AUC was 3.9-fold higher. Dose restrictions are recommended in patients with mild and moderate hepatic impairment. Pitavastatin is contraindicated in patients with severe hepatic impairment.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lactose Monohydrate, Crospovidone, Hypromellose 3 CPS 2910, Sodium Carbonate Anhydrous, Calcium Carbonate, Magnesium Stearate and Opadry White 03F58750 (Hypromellose, Macrogol, Talc and Titanium Dioxide).

6.2 Incompatibilities

Not applicable

6.3 Shelf life

Refer to details on outer packaging.

6.4 Special precautions for storage

Do not store above 30°C. Protect from light and moisture. Keep out of reach of children.

6.5 Nature and contents of container

Packed in Alu-Alu blister pack (a blister formed from 0.025 mm Aluminum foil and Cold forming foil 25µOPA/45µAlu/60µPVC) of 10's. Such 3 blisters are packed in one printed carton with pack insert.

6.6 Special precautions for disposal

To protect the environment, do not dispose of via waste water or household waste.

7. MANUFACTURER'S NAME AND ADDRESS



Zydus Lifesciences Limited,
Kundaim Industrial Estate,
Plot No. 203-213, Kundaim,
Goa - 403 115, INDIA

8. DATE OF REVISION OF THE TEXT

04 March 2026

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